# New Embeddings That Fixed the Wrong Problem: Experiments 08, 09 and 10 Rerun

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## The problem we started from

Report 03 ended with one measurement: all the patient embeddings looked almost identical to each other. Two random patients had a similarity of .979 out of 1, so the "20 nearest neighbours" in the graph were chosen by differences in the fourth decimal place.

The lab proposed a set of new encoders to fix this. This report describes them and what happened when experiments 08, 09 and 10 were run again on top of them.

## The new embeddings

The old encoder was trained on one thing only: predict the label. Nothing stopped it from throwing away every direction in the embedding except the one that helps with that prediction, and that is what it did. The new recipes add a second term to the training objective that pushes back: each of the 64 numbers in an embedding must keep some spread, and different numbers must not repeat each other.

Five versions were trained. Each is scored two ways: how similar two random patients look (lower is better spread), and how well the embeddings predict the label on their own, before any graph is involved.

| Version | What it does | Similarity | AUROC |
|---|---|---|---|
| A | The old recipe, reproduced | .981 | .798 |
| B | A bigger encoder, same objective | .890 | .769 |
| C | Old encoder + the anti-collapse term | **.633** | **.788** |
| D | Bigger encoder + the anti-collapse term | .356 | .742 |
| E | Uses the patient's real visit sequence + the term | .338 | .710 |

Table 1: The five encoders, one run each. Similarity is between two random patients.

There is a clear trade: the more the embeddings are forced to spread out, the worse they predict the label. Version C is the compromise — it removes most of the crowding while giving up only .010 of prediction quality — so C is the one taken forward.

The fix works on its own terms. Random patients moved from .979 to .633, and the similarities among the neighbour candidates, which is what actually decides the graph, went from a range of .009 to a range of .080. The neighbours are now chosen by real differences instead of rounding noise.

| Model | Old embeddings | Version C |
|---|---|---|
| rgat | **.800** | .773 |
| rgcn | .784 | .768 |

Table 2: Test AUC on the color task. Best setting chosen on validation, one run each.

## Experiment 08 again

**Everything got worse.** Both models lost between .016 and .027. Repeated runs of the same setting differ by about .004–.009, so these drops are three to seven times larger than the normal noise.

**It is not bad luck in the tuning.** The whole set of tried settings moved down, not just the best one: the middle setting went from .783 to .770. Validation scores dropped by the same amount.

**It is missing information, not overfitting.** The score on the *training* data fell as well, from .866 to .803. The model is not memorising more and generalising less; there is simply less to work with.

## Experiment 09 again

| Readout | Old embeddings | Version C |
|---|---|---|
| `supernode` | $.795 \pm .009$ | $\mathbf{.780 \pm .008}$ |
| `pma` | $.795 \pm .004$ | $.776 \pm .007$ |
| `gated_attn` | $.796 \pm .005$ | $.773 \pm .004$ |
| `node_score_mean` | $.796 \pm .002$ | $.772 \pm .004$ |
| `mean` (the old method) | $\mathbf{.798 \pm .004}$ | $.769 \pm .010$ |
| `multistat` | $.792 \pm .007$ | $.768 \pm .010$ |

Table 3: Test AUC on the color task, 5 runs each.

**The order changed.** On the old embeddings nothing beat the plain average and the six numbers were indistinguishable. On the new ones the plain average drops to fifth place, and `supernode` — an extra node per patient group that gathers the information — is now .011 ahead of it. On the per-visit task the same gap is .022.

**But this is a shuffle inside a lower range.** The best readout on the new embeddings (.780) is still below the *worst* readout on the old ones (.792).

## Experiment 10 again

**The flexible graph finally does something.** On the old embeddings it was .004 ahead of the fixed graph, small enough to be explained by the extra tuning it received. On the new embeddings the gap is .013, about three times the run-to-run noise.

| Graph | Old embeddings | Version C |
|---|---|---|
| Flexible | $\mathbf{.800 \pm .003}$ | $\mathbf{.780 \pm .005}$ |
| 20 neighbours (the old graph) | $.796 \pm .002$ | $.767 \pm .004$ |

Table 4: Test AUC on the color task, rgat, 5 runs each.

**And it now prefers the opposite settings.** On the old embeddings the winner was a dense graph (63 connections per patient) built on the untouched similarities. On the new ones the winner is a sparse graph (12 connections) built on whitened similarities, and whitening is now the best of the three ways of measuring similarity (.794, against .780 untouched and .770 centred). The choice of similarity measure used to make no difference at all.

**The gain does not close the gap.** The best result on the new embeddings is $.780 \pm .005$, against $.800 \pm .003$ on the old ones.

## What happens with no embeddings at all

Experiments 08, 09 and 10 were run once more with the embeddings taken out of the model's input. The graph was still built from them, so the same visits were still neighbours. Only the vector each visit starts from changed. Four starting vectors were compared. This ran on the old embeddings, so the number to compare with is .800.

| What each visit starts from | Visits start | Test AUC | On training data |
|---|---|---|---|
| Its embedding: the 64 numbers the encoder made from its text (every result above) | different, by text | **.800** | .851 |
| A random vector, drawn once before training and never changed | different, by noise | .702 | .831 |
| A vector of its own that the model is free to change while it trains | different, by noise | .701 | 1.000 |
| One single vector, shared by every visit in the dataset | identical | .499 | .504 |

Table 5: Experiment 08 rerun with the embeddings removed from the input. All four rows use the same graph, so in all four the same visits are neighbours; only the starting vectors differ. Test AUC on the color task, rgat, best setting chosen on validation, one run each. Chance is .500.

**Without the embeddings the score falls from .800 to .70.** Chance is .500.

**A trained vector and a random one score the same** — .701 and .702. The trained one scores a perfect 1.000 on the training data, so it is memorising the visits it has seen.

**One shared vector scores chance.** Every layer of the network averages over neighbours, so visits that start the same stay the same.

**The choice of readout starts to matter.** The six readouts spread over .057 without the embeddings, against .006 with them.

| | With embeddings | Without |
|---|---|---|
| Exp. 09: best of the six readouts | $.798 \pm .004$ (`mean`) | $.679 \pm .029$ (`pma`) |
| Exp. 09: worst of the six | $.792 \pm .007$ (`multistat`) | $.622 \pm .017$ (`supernode`) |
| Exp. 10: flexible graph | $.800 \pm .003$ | $.705 \pm .020$ |
| Exp. 10: fixed graph, 20 neighbours each | $.796 \pm .002$ | $.670 \pm .029$ |

Table 6: The left column repeats tables 3 and 4; the right column is the same experiment with the embeddings taken out of the input, using the trained vector from row 3 of table 5. The two top rows are the best and the worst of the six readouts of experiment 09 — which readout that is differs between the columns, so each one is named. The two bottom rows are the two graphs of experiment 10. What the text is about is the size of the gap between the rows of a pair, not the rows themselves. Test AUC on the color task, rgat, 5 runs each.

**So does the choice of graph.** The flexible graph beats the fixed one by $.035$ without the embeddings, against $.004$ with them.

**The winning graph is the same kind as before** — dense, $55$ connections per patient, built on the untouched similarities.

## What this adds up to

**The crowding was not the bottleneck.** It was a real defect and it has been fixed, but fixing it made every downstream number worse. The one direction the old encoder had concentrated on was carrying the signal; spreading the embeddings out spent it.

**The graph and the readout do respond now.** Both parts of the pipeline that looked dead in report 03 came back to life on the new embeddings: the flexible graph, the choice of similarity measure, and the choice of readout all matter where none of them did before. This says those components were never broken — they had nothing to work with.

**The two effects do not cancel out.** The parts we can tune gained about $.013$; the encoder change cost about $.030$. The best model overall is still the old one, at $.800 \pm .003$.